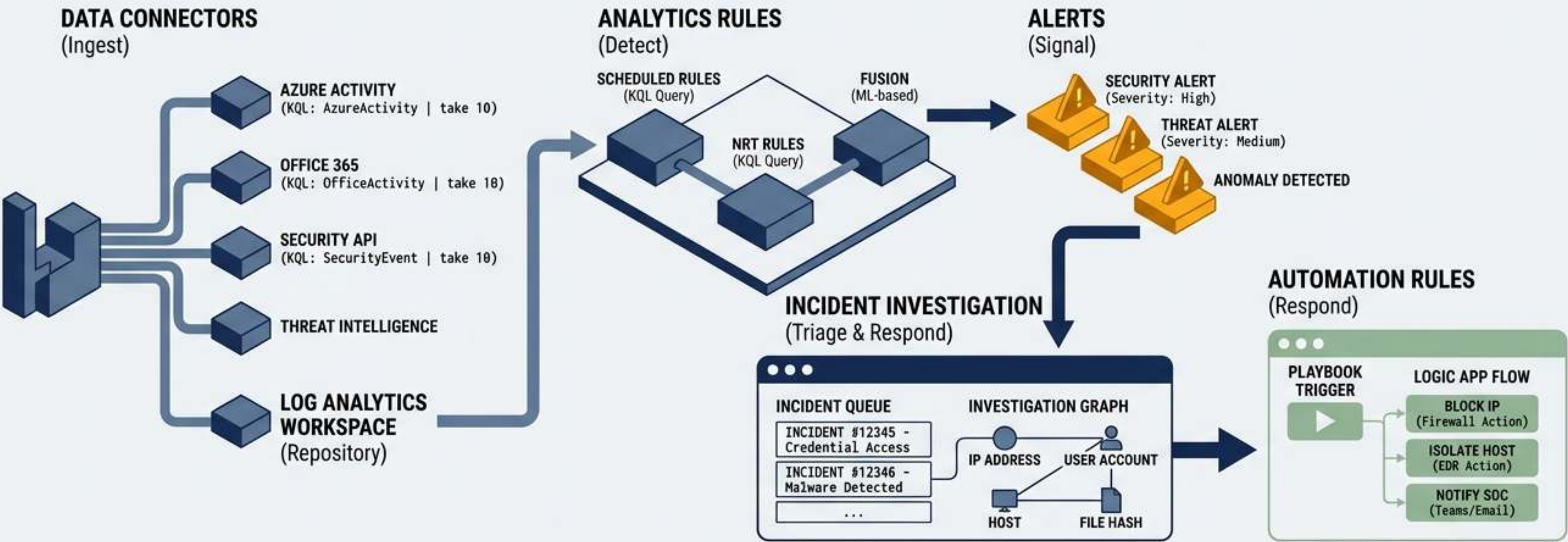


MICROSOFT SENTINEL PIPELINE AND ARCHITECTURE

DATA CONNECTORS, ANALYTICS RULES & INCIDENT INVESTIGATION



LEGEND:
[NAVY] STRUCTURAL / PRIMARY COMPONENTS
[SLATE BLUE] SECONDARY / DATA FLOW
[AMBER] ALERTS / INCIDENTS
[GREEN] AUTOMATED RESPONSES / SUCCESSFUL FLOWS
[TYPOGRAPHY] SANS-SERIF FOR HEADERS, MONOSPACED FOR TECHNICAL DETAILS/KQL

THE INTELLIGENCE LAYER OVER LOG ANALYTICS

TOP LAYER

MICROSOFT SENTINEL

Cloud-native SIEM + SOAR.

Does not store data separately.

Provides the analytics, incidents, and automation on top of the workspace.



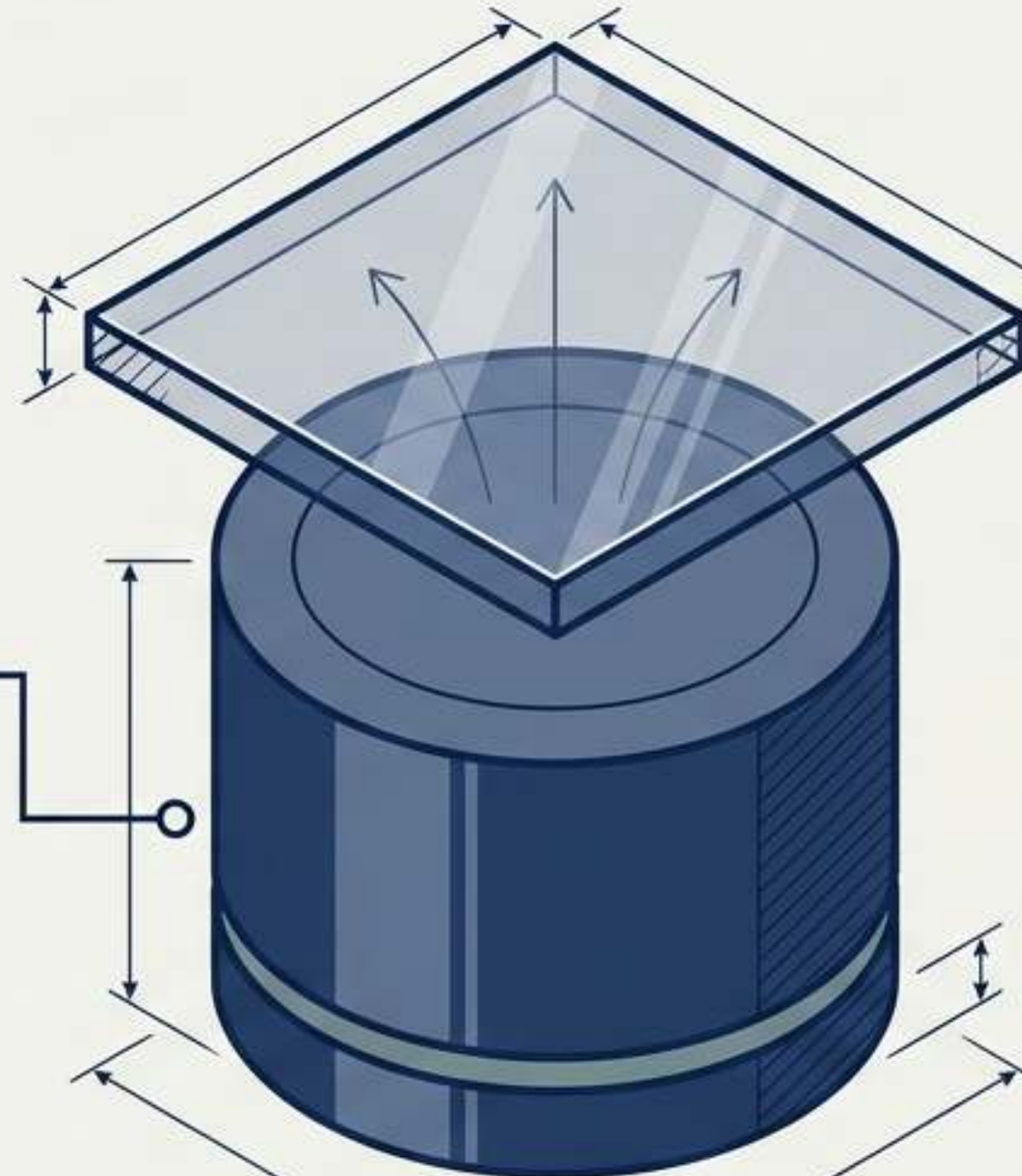
BOTTOM LAYER

LOG ANALYTICS WORKSPACE

The foundation.

Stores the actual log data.

Region choice is permanent and critical for compliance.



EXAM NOTE:

To enable Sentinel, you need Contributor on the subscription.

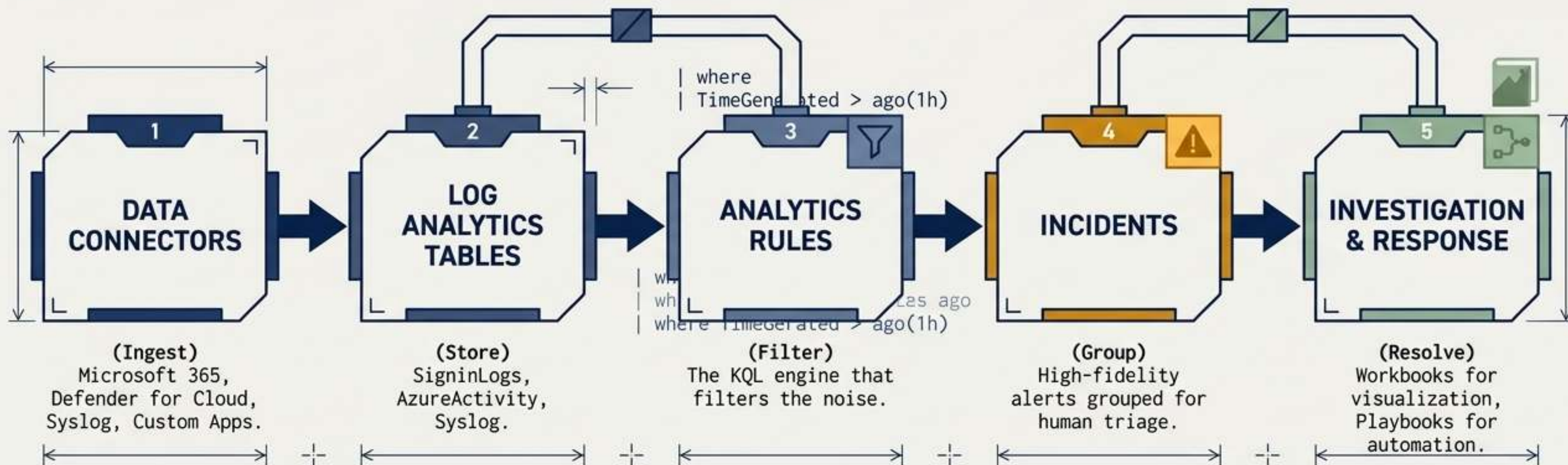
To use it, Contributor or Reader on the Resource Group is sufficient.



PROJECT:
AZURE SECURITY ARCHITECTURE
BRANCH: TSTLA:
SENTINEL LAYER CAKE

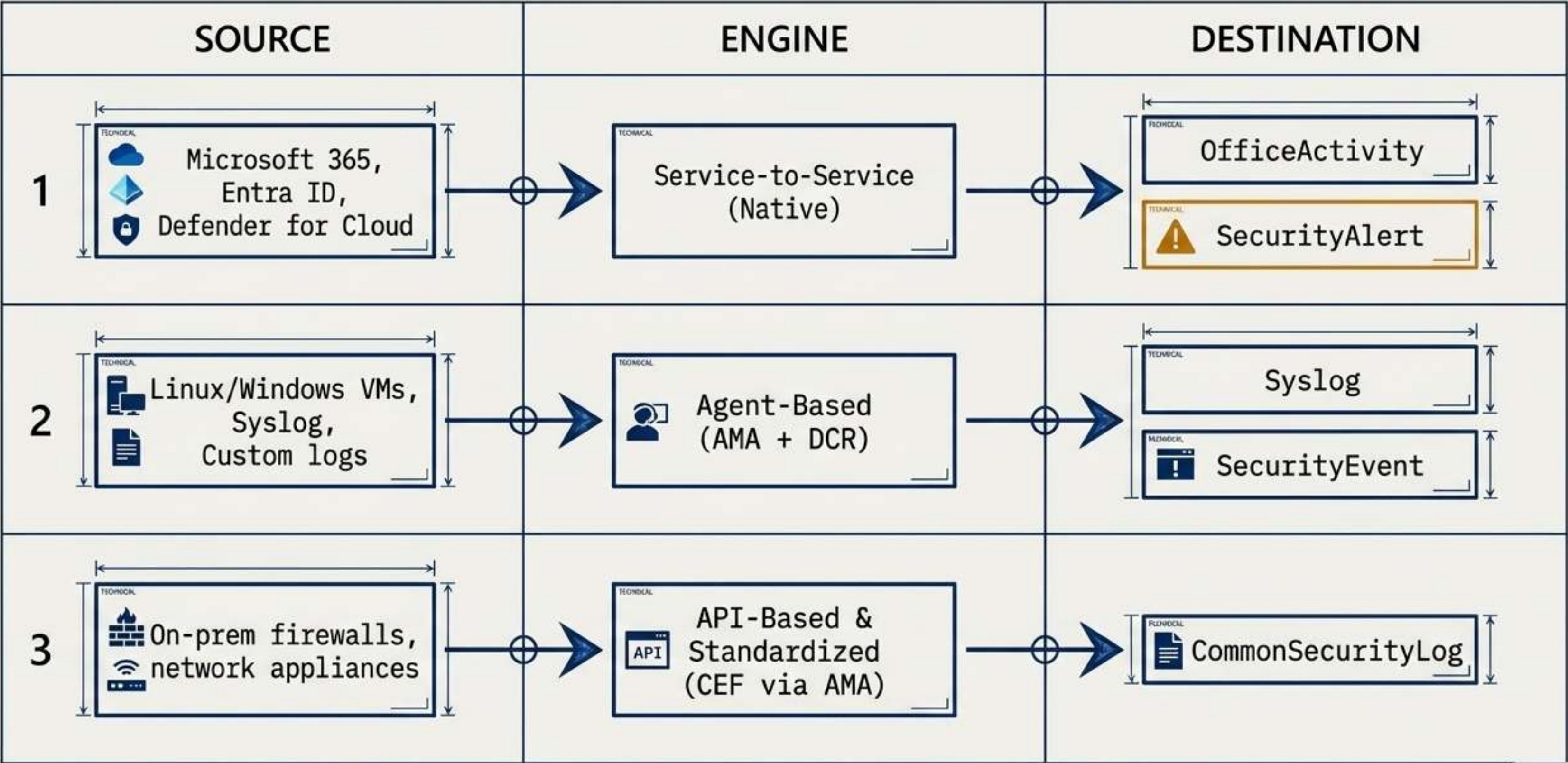
SCALE:
NTS
DATE:
OCTOBER 26, 2023

THE SENTINEL DATA ASSEMBLY LINE



TAKEAWAY: You cannot have an **Analytics Rule** without ingested **Data**.
You cannot trigger a **Playbook** without an **Incident**.

BRINGING TELEMETRY INTO THE WORKSPACE



CONNECTOR NUANCES AND EXAM MECHANICS

Entra ID Granularity

Entra ID logs write to multiple distinct tables (SignInLogs, AuditLogs, etc.).

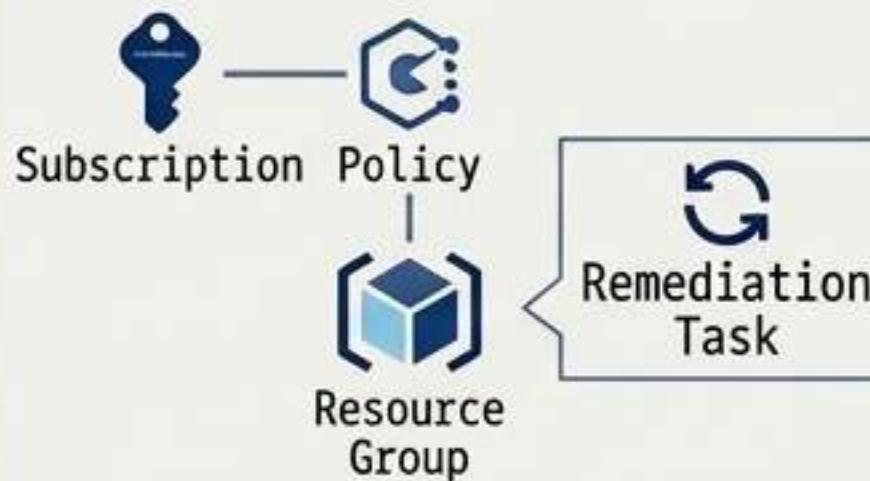
Enable only what you need to control ingestion costs.

- ☒ SignInLogs
- ☒ AuditLogs
- ☐ ...
- ☒ ManagedIdentitySignInLogs
- ☒ ADFSSignInLogs
- ☐ ...

Azure Activity Policy

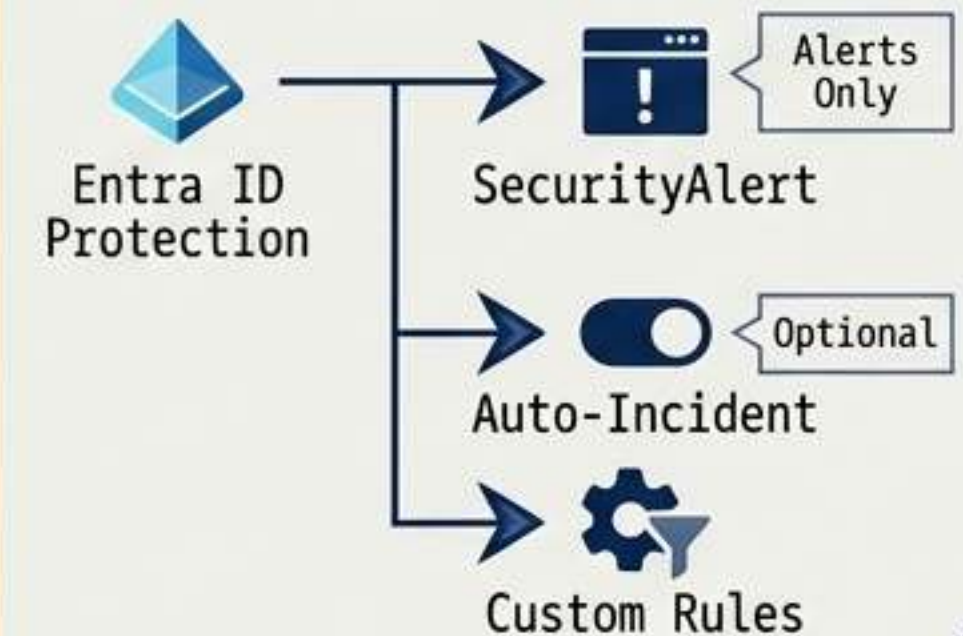
Azure Activity is the odd one out. It is deployed via an Azure Policy assignment at the subscription level.

You must create a remediation task to capture existing resources.



Entra ID Protection

Streams identity risk alerts into the SecurityAlert table, but not the raw forensic evidence. Auto-incident creation is optional; custom rules offer better control.



ANATOMY OF A SCHEDULED ANALYTICS RULE

THE TRIGGER

Condition to generate an alert.
E.g., Results > 0.
Threshold-based activation.

THE GROUPEUR

Incident settings.
Groups multiple triggered alerts into a single actionable incident to prevent SOC fatigue.
Reduces alert noise, improves efficiency.

THE ENGINE

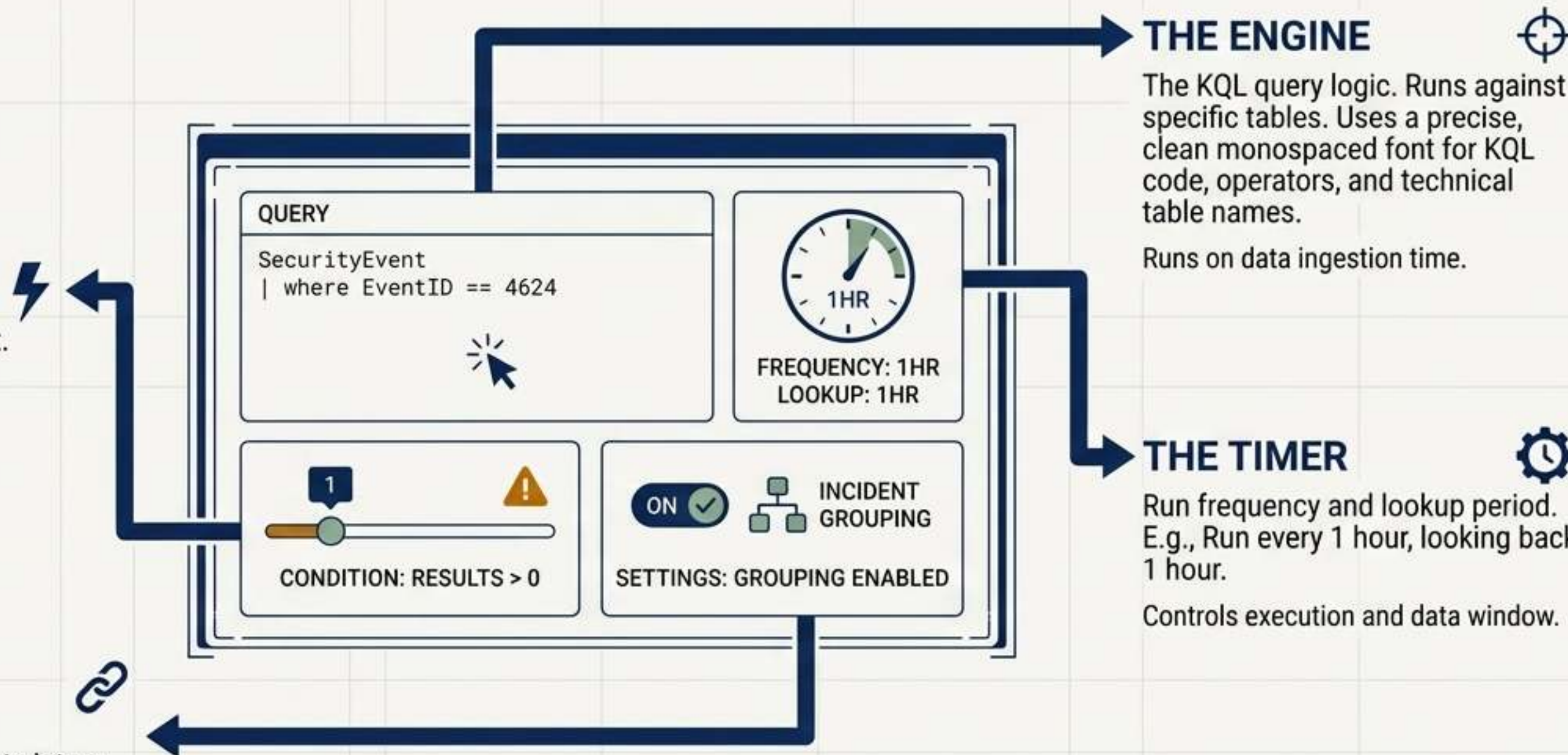
The KQL query logic. Runs against specific tables. Uses a precise, clean monospaced font for KQL code, operators, and technical table names.

Runs on data ingestion time.

THE TIMER

Run frequency and lookup period.
E.g., Run every 1 hour, looking back 1 hour.

Controls execution and data window.



KQL AS A FILTRATION SEQUENCE

KQL CODE

PLAIN ENGLISH TRANSLATION

SigninLogs	→	Look inside the Entra ID Sign-in logs table.
where ResultType == '50126'	→	Filter for this specific 'Invalid Password' error code.
summarize count() by bin(TimeGenerated, 1h), IPAddress	→	Group the results into 1-hour buckets, and count the failures per IP address.

 **AZ-500 EXPECTATION:** Read the query to identify the correct detection logic. You do not need to write it from scratch.

OPERATOR ROSETTA STONE

KQL OPERATOR	FUNCTIONAL EXPLANATION
where	Filters the noise. Only show me records that match this exact condition.
summarize	Aggregates the data. Count, average, or find the max value of these events.
bin	Time-buckets data. Group these timestamps into 1-hour or 1-day chunks.
join	Connects two tables. Match a user ID from SigninLogs with their role in IdentityInfo.
project	Selects specific columns. Throw away the rest of the data and only show me the IP and Username.

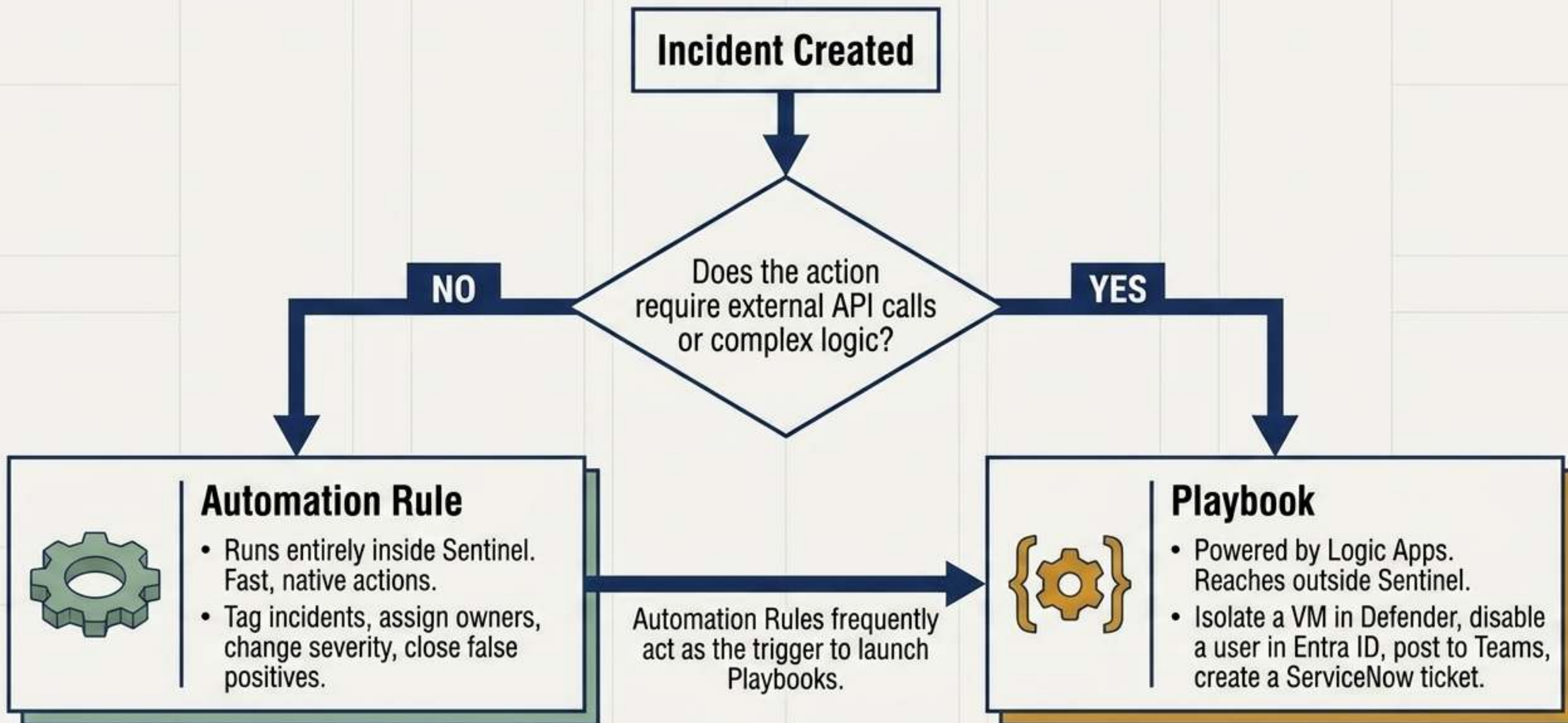
KQL operators function as commands within a query sequence, building upon the results of the preceding operator.

THE INCIDENT INVESTIGATION WORKSPACE



An **Incident** is the primary unit of investigation. It groups related alerts, maps entities across different log sources, tracks severity, and moves through a status lifecycle (New -> Active -> Closed).

RESOLVING THREATS AT MACHINE SPEED



ANALYTICS RULE ENGINES COMPARED

RULE TYPE	TRIGGER FREQUENCY	BEST USE CASE	LOGIC SOURCE
 Scheduled	Interval-based (e.g., 1hr)	Most common custom detections	KQL Query
 Near-Real-Time (NRT)	Every 1 minute	 Time-sensitive threats (impossible travel)	KQL Query
 Fusion	 Continuous	High-confidence correlation of low-fidelity signals	Built-in Machine Learning (No KQL needed)
 Microsoft Security	 Continuous	Auto-promoting Defender alerts to Incidents	Native Integration

SCENARIO: IMPOSSIBLE TRAVEL DETECTION

THE THREAT



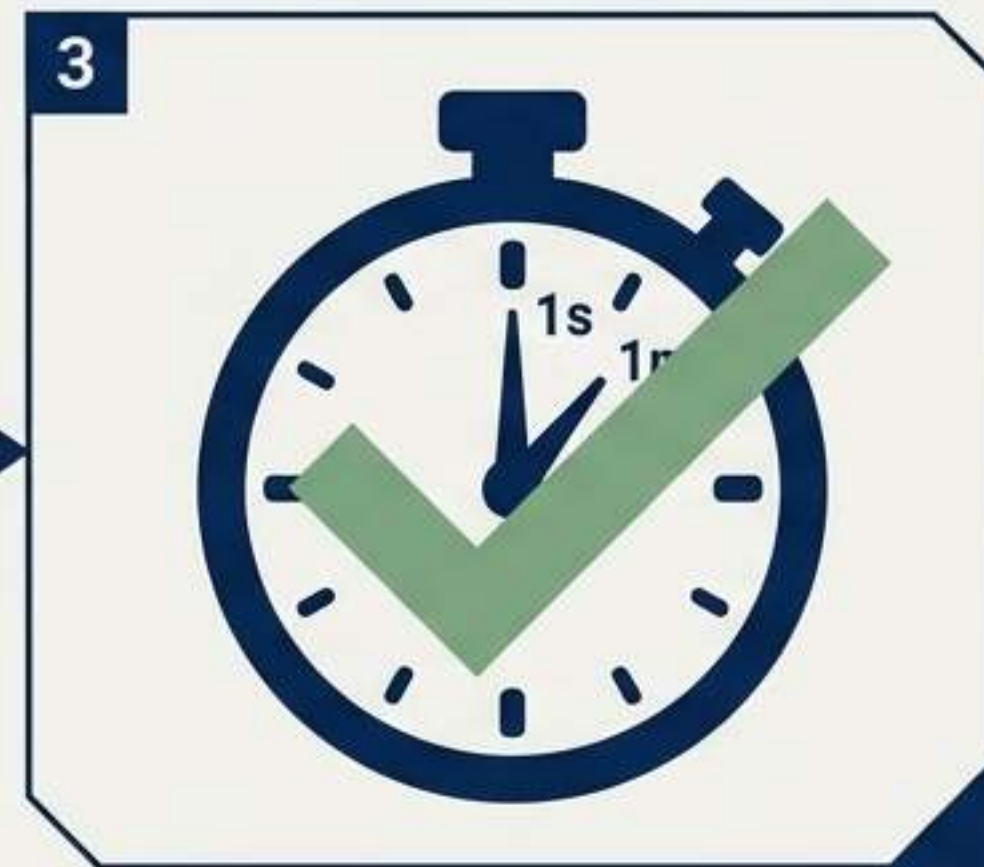
A user logs in from New York, then 5 minutes later, logs in from Tokyo.

THE FAILURE



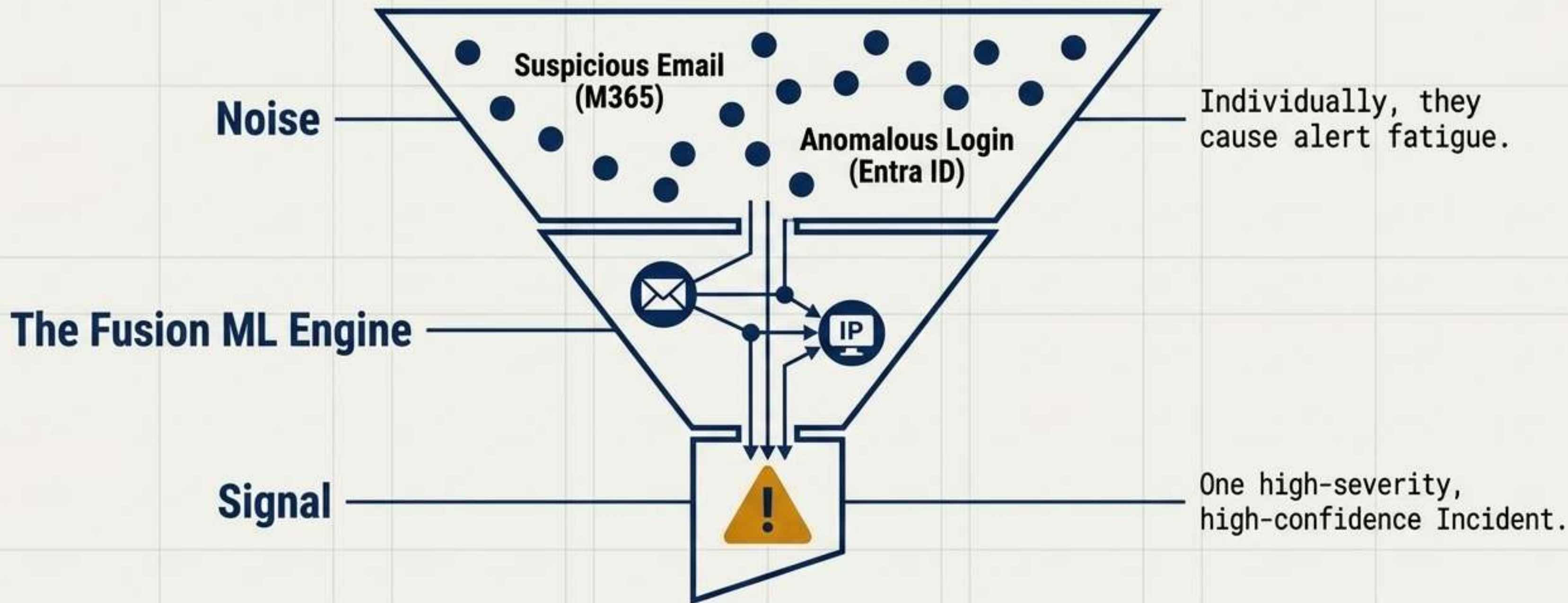
A standard **Scheduled Rule** running every 4 hours won't catch this until the attacker has already exfiltrated data.

THE SOLUTION



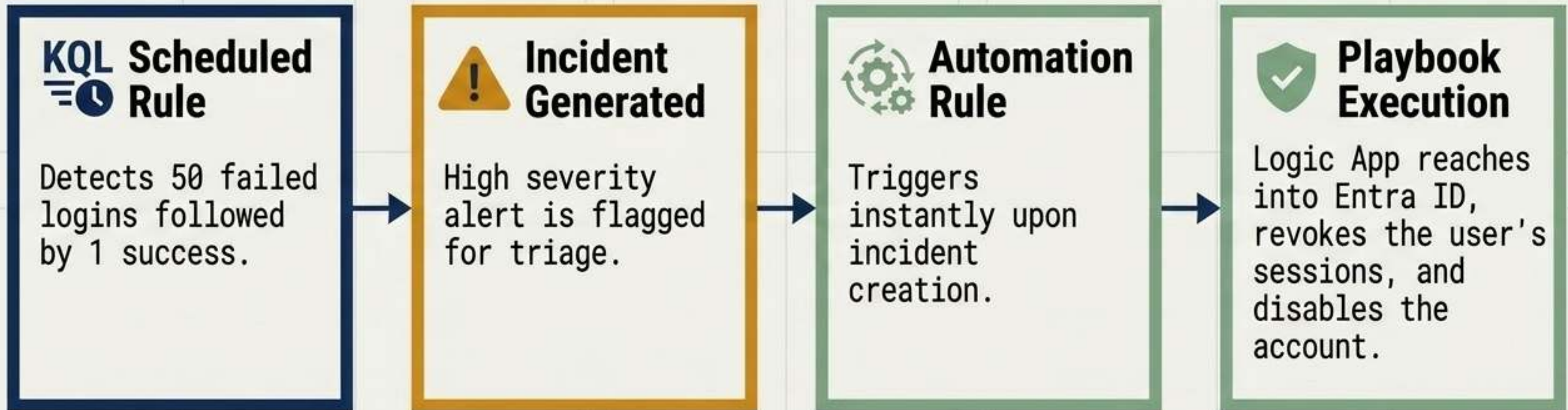
Near-Real-Time (NRT) Rule. Runs KQL every 1 minute with a 1-minute lookback. Detects the impossible travel instantly and raises an incident before damage occurs.

SCENARIO: LOW-FIDELITY SIGNAL CORRELATION



Takeaway: Fusion uses built-in Machine Learning to correlate multi-stage attacks across different kill-chain phases. No KQL authoring required.

SCENARIO: AUTOMATIC ACCOUNT CONTAINMENT



Takeaway: The identity of the Logic App requires the Microsoft Sentinel Responder role to read/update the incident and specific Entra ID permissions to disable the user.

EXAM TRAPS & COMMON MISCONCEPTIONS



Trap: Confusing Workbooks with Analytics Rules.

Workbooks are for passive visual dashboards and trend analysis only.



Fact: Analytics Rules create incidents and

Analytics Rules create incidents and trigger alerts. Workbooks are for passive visual dashboards and trend analysis only.



Trap: Assuming Basic Log tables can be used for complex hunting.

Basic Logs are cheap but do not support join or summarize in KQL.



Fact: Basic Logs are cheap but do not

Basic Logs are cheap but do not support join or summarize in KQL.



Trap: Applying an Azure Policy for Azure Activity logs without a remediation task.

Policy only applies to new resources. The remediation task is strictly required to



Fact: Policy only applies to new

Policy only applies to new resources. The remediation task is strictly required to capture existing subscriptions.

Takeaway: Master the distinctions between tools and their specific purposes to avoid common pitfalls during the exam. Pay close attention to KQL capabilities and policy scopes.

THE COMPLETE SENTINEL ARCHITECTURE

Master the flow of data, and you master the architecture.

